

POST IMAGE PROCESSING IN ULTRASOUND SIMULATION

Module of the NITK-USoundSim Project

Seminar Report — Course Code: EC787

A Project Report submitted in partial fulfilment of the
requirements for the degree of

MASTER OF TECHNOLOGY

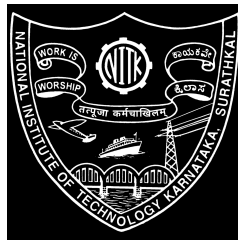
in

SIGNAL PROCESSING AND MACHINE LEARNING

submitted by

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DECLARATION

I hereby declare that the report of the project work titled “**Post Image Processing in Ultrasound Simulation**”, which is being submitted to the National Institute of Technology Karnataka, Surathkal, in partial fulfilment of the requirements for the award of the Degree of **Master of Technology** in the Department of Electronics and Communication Engineering, is a bonafide report of the work carried out by me. The material contained in this report has not been submitted to any University or Institution for the award of any degree.

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CERTIFICATE

This is to certify that the project report titled “**Post Image Processing in Ultrasound Simulation**” submitted by **Darshil Maniya** (Register Number: 262SP009) as the record of the work carried out by him, is accepted as the *Project Work Report* submission in partial fulfilment of the requirements for the award of the degree of **Master of Technology** in the Department of Electronics and Communication Engineering, National Institute of Technology Karnataka, Surathkal.

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ABSTRACT

Ultrasound is among the most widely used diagnostic imaging modalities because it is inexpensive, portable, real-time and free of ionising radiation. Its principal image-quality limitation is *speckle* — a granular interference pattern produced when echoes from sub-resolution scatterers add coherently at the transducer. Speckle is deterministic rather than random electronic noise, it is present in every scan, and it obscures low-contrast lesions, blurs organ boundaries and degrades the performance of automated analysis methods.

This report describes the **Post Image Processing** module of **NITK-UsoundSim**, a software ultrasound imaging system developed by a sixteen-member student team across eight modules. This module is the terminal stage: it accepts the log-compressed, scan-converted B-mode image from the B-Mode Image Formation module and returns a display-ready image in which speckle has been suppressed without destroying diagnostically relevant structure. A four-stage chain is proposed and implemented — depth-gain normalisation, despeckle filtering, contrast and edge restoration, and automated quality assessment. Five classical filters (median, Lee, Speckle Reducing Anisotropic Diffusion, Non-Local Means and wavelet soft thresholding) are implemented in Python and compared quantitatively; no machine-learning model is used at any stage.

Because the upstream dependency was unavailable during development, a synthetic B-mode testbench was constructed for validation. It generates a sector-format phantom containing an anechoic cyst, a hyperechoic lesion, a hypoechoic nodule and point targets with fully developed Rayleigh speckle, and supplies a noise-free ground-truth image permitting reference-based metrics.

On this testbench, SRAD achieved the best overall balance, raising the contrast-to-noise ratio from 1.78 to 2.98 and the speckle signal-to-noise ratio from 4.28 to 9.29 while retaining an edge-preservation index of 0.96. Non-Local Means attained marginally higher noise suppression (SNR 9.77) but retained only 0.81 of the edge information and was the slowest method evaluated. SRAD is therefore recommended as the default filter for the module.

Keywords: Ultrasound imaging, B-mode, speckle, despeckling, anisotropic diffusion, SRAD, non-local means, contrast-to-noise ratio, image enhancement.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
B-mode	Brightness Mode
CLAHE	Contrast Limited Adaptive Histogram Equalisation
CNR	Contrast-to-Noise Ratio
DGC	Depth Gain Compensation
EPI	Edge Preservation Index
FWHM	Full Width at Half Maximum
NLM	Non-Local Means
PSF	Point Spread Function
PSNR	Peak Signal-to-Noise Ratio
RF	Radio Frequency
ROI	Region of Interest
SNR	Signal-to-Noise Ratio
SRAD	Speckle Reducing Anisotropic Diffusion
SSIM	Structural Similarity Index Measure
TGC	Time Gain Compensation
VCT	Virtual Clinical Trial

CHAPTER 1

INTRODUCTION AND BACKGROUND

1.1 Background

Medical ultrasound imaging uses high-frequency acoustic pulses, typically 2–15 MHz, to visualise internal anatomy. A transducer array on the skin transmits a focused pulse; a fraction of its energy is reflected at every interface where acoustic impedance changes. The returning echoes are beamformed, envelope-detected, log-compressed and mapped to grey levels to form the brightness-mode (B-mode) image [1].

Ultrasound uses no ionising radiation, costs a small fraction of a computed tomography or magnetic resonance system, is portable enough for bedside use, and produces images in real time. These properties make it the highest-volume diagnostic imaging modality worldwide.

1.2 Motivation

Optimising imaging systems through physical experimentation is slow and expensive, and clinical trials are constrained by ethics, cost and the absence of ground truth. Abadi *et al.* [2] argue that *virtual clinical trials*, in which patient, scanner and interpreter are all replaced by computational models, offer a practical alternative. A software ultrasound simulator is such a virtual scanner: transducer geometries, transmit sequences, beamforming strategies and image-processing algorithms can all be evaluated against a known ground truth at negligible cost.

The **NITK-UsoundSim** project builds such a simulator. Sixteen students are organised into eight two-member module teams spanning the complete acoustic and signal-processing chain. The present report documents the final module of that chain: **Post Image Processing**.

1.3 Problem Statement

Every B-mode image contains *speckle*. It is not additive electronic noise but arises from coherent summation of echoes returned by scatterers smaller than the system resolution [3]: hundreds of such scatterers within a resolution cell contribute echoes of essentially random phase, which sum constructively at some positions and destructively at others. Because this is a consequence of coherent imaging physics, it cannot be removed by improving the scanner electronics. Its effects on diagnostic quality are threefold:

1. **Reduced lesion conspicuity.** Low-contrast lesions whose echogenicity differs only slightly from surrounding tissue are masked by the speckle pattern and may be missed entirely.
2. **Boundary uncertainty.** Organ and lesion boundaries become indistinct, so measurements of size, volume and wall thickness carry large uncertainty and vary between observers.

3. **Degraded automated analysis.** Segmentation, registration and computer-aided detection algorithms perform poorly on images dominated by a high-variance texture.

This report therefore designs, implements and validates a post-processing stage that suppresses speckle while preserving diagnostically valuable structure. The two requirements are in direct opposition: any operator strong enough to remove the granular texture can also remove genuine small structures. Managing this trade-off quantitatively is the central engineering problem of this work.

1.4 Objectives

The specific objectives of the Post Image Processing module are:

1. To define a clear data interface with the upstream B-Mode Image Formation module, specifying array format, dynamic range, geometry and accompanying metadata.
2. To design a modular post-processing pipeline covering brightness normalisation, speckle suppression, contrast restoration and quality assessment.
3. To implement and compare five classical despeckling filters using only deterministic image-processing techniques, without any machine-learning component.
4. To establish an objective evaluation framework based on contrast-to-noise ratio, speckle signal-to-noise ratio and edge preservation, supplemented by reference-based metrics where ground truth is available.
5. To recommend a default filter and parameter set for integration into the NITK-UsoundSim system, supported by measured evidence.

1.5 Scope and Limitations

The module begins at the log-compressed, scan-converted B-mode image and ends at the display-ready image. All upstream acoustic and beamforming stages belong to other module teams and are treated here only as far as necessary to explain the input.

Two limitations are stated at the outset. Only classical image-processing operators are considered; learning-based despeckling is discussed in Chapter 4 as future work. Second, the upstream module had not delivered production output at the time of writing, so validation used the synthetic testbench of Chapter 2.8; all results in Chapter 3 will require re-tuning against production data.

1.6 Organisation of the Report

Chapter 1.7 reviews the B-mode chain, the origin of speckle and prior despeckling work. Chapter 2 presents the pipeline, the data interface and the filter formulations. Chapter 2.8 covers the software architecture, testbench and protocol. Chapter 3 reports the results, and Chapter 4 concludes.

1.7 Principle of B-Mode Imaging

A transducer in contact with the skin emits a short acoustic pulse; echoes returning from tissue interfaces are received by the same array, with amplitude mapped to brightness and arrival time to depth.

The Signal Processing Chain

The transformation from channel data to displayed image follows the well-established sequence of Figure 1.1 [1].

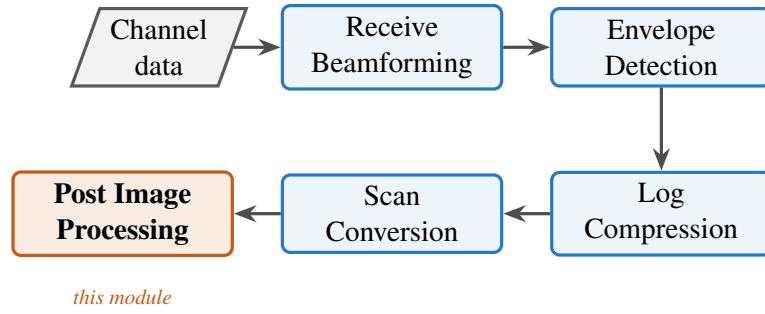


Figure 1.1: The B-mode signal-processing chain. The present work implements the final stage, shown highlighted.

Receive beamforming applies channel-dependent delays and apodisation weights so echoes from a chosen focal point add coherently, giving one radio-frequency (RF) line per beam direction. **Envelope detection** then extracts the amplitude modulation: if $s(t)$ is the RF signal and $\hat{s}(t)$ its Hilbert transform, the analytic signal is $s_a(t) = s(t) + j\hat{s}(t)$ and

$$E(t) = |s_a(t)| = \sqrt{s^2(t) + \hat{s}^2(t)}. \quad (1.1)$$

Logarithmic compression maps the wide dynamic range of E (commonly 60–100 dB) into the 8-bit range a display can render:

$$I_{\text{dB}} = 20 \log_{10} \left(\frac{E}{E_{\text{max}}} \right), \quad I = 255 \cdot \text{clip} \left(\frac{I_{\text{dB}} + D}{D}, 0, 1 \right), \quad (1.2)$$

where D is the displayed dynamic range in decibels.

Scan conversion finally resamples the polar data (r, θ) onto the Cartesian display grid using $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(x/y)$, with bilinear interpolation between the four nearest samples, producing the characteristic fan-shaped sector image.

1.8 The Physics and Statistics of Speckle

Physical Origin

Within any resolution cell there are many scattering structures — cell clusters, fibres, small vessels — far smaller than the pulse wavelength. Each returns an echo whose phase depends on its position, and the received signal is their coherent sum (Figure 1.2).

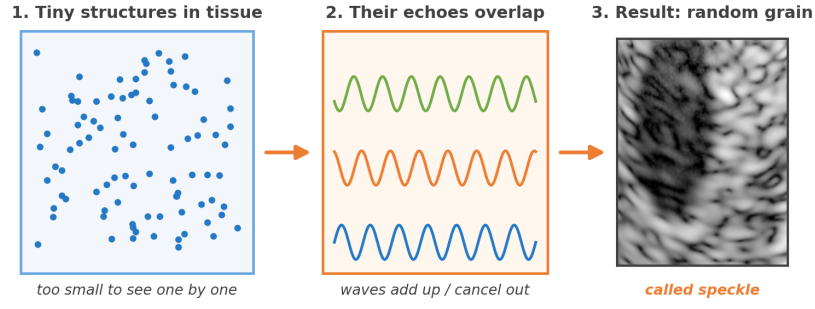


Figure 1.2: Formation of speckle. Sub-resolution scatterers (left) each return an echo; these echoes interfere constructively and destructively (centre), producing the granular pattern observed in the image (right).

Statistical Model

For *fully developed speckle* — many scatterers per cell, none dominant — the real and imaginary parts of the complex signal are independent zero-mean Gaussian variables, so the envelope E is Rayleigh distributed [4]:

$$p(E) = \frac{E}{\sigma^2} \exp\left(-\frac{E^2}{2\sigma^2}\right), \quad E \geq 0. \quad (1.3)$$

A consequence is that the ratio of mean to standard deviation is constant, independent of tissue echogenicity:

$$\text{SNR}_{\text{speckle}} = \frac{\mu_E}{\sigma_E} = \frac{\sqrt{\pi/2}}{\sqrt{2 - \pi/2}} \approx 1.91. \quad (1.4)$$

Equation (1.4) matters in practice: a measured value near 1.91 in a homogeneous region confirms fully developed speckle, while a substantially higher value indicates smoothing. It therefore verifies testbench realism and provides a natural despeckling metric.

Before log compression speckle is *multiplicative*: the envelope is the product of tissue reflectivity R and a unit-mean speckle field n ,

$$E(x, y) = R(x, y) \cdot n(x, y). \quad (1.5)$$

Applying a logarithm converts this product into a sum,

$$\log E = \log R + \log n, \quad (1.6)$$

the basis of *homomorphic* filtering: additive-noise techniques such as wavelet thresholding may be applied in the log domain and the result exponentiated back.

Why Speckle Should Not Be Eliminated Entirely

Speckle carries information: its texture statistics support tissue characterisation, its inter-frame correlation drives speckle-tracking elastography, and the granular appearance of an organ is itself a diagnostic cue. Complete removal would yield a cartoon-like image and break downstream tracking. The objective is *suppression with structure preservation*, not elimination.

1.9 Review of Despeckling Methods

Local Statistics Filters

Adaptive local-statistics filters estimate the mean and variance in a sliding window and interpolate between the raw pixel and the local mean according to neighbourhood homogeneity. Lee [5] introduced this for multiplicative noise and Kuan *et al.* [6] generalised it to signal-dependent noise; the Frost filter substitutes an exponentially decaying kernel. These remain a cheap, standard baseline.

Anisotropic Diffusion

Perona and Malik [7] treated denoising as diffusion whose conduction coefficient decreases with local gradient magnitude, inhibiting diffusion across edges. Yu and Acton [8] adapted this to multiplicative speckle as Speckle Reducing Anisotropic Diffusion (SRAD), replacing the gradient with an instantaneous coefficient of variation. SRAD is the reference classical method for ultrasound.

Non-Local and Transform-Domain Methods

Buades *et al.* [9] exploited pattern repetition in natural images, weighting pixels by the similarity of their surrounding *patches* rather than by spatial proximity; Non-Local Means (NLM) achieves excellent suppression in homogeneous regions at high cost. Donoho [10] showed soft thresholding of wavelet coefficients to be near-optimal for additive noise, which with Equation (1.6) yields wavelet despeckling methods that preserve texture but suppress less speckle than diffusion [11].

Five representative operators were selected for implementation: the median filter (order statistic), Lee (local statistics), SRAD (diffusion), NLM (non-local) and wavelet soft thresholding (transform domain). Total-variation and learning-based methods were excluded — the former as redundant with SRAD, the latter as requiring paired training data outside this scope.

The literature offers many despeckling operators but few side-by-side evaluations on a common dataset with known ground truth. Simulation-based evaluation fills this gap [2]: a virtual pipeline supplies noise-free references against which algorithms can be scored objectively. This module compares five filters under identical conditions using reference-based and reference-free metrics.

CHAPTER 2

METHODOLOGY AND IMPLEMENTATION

2.1 Position of the Module in the System

NITK-USoundSim is decomposed into eight modules, each assigned to a two-member team: Transducer, Transmit Beamforming, Acoustic Propagation, Tissue Interaction, Receiver Beamforming, B-Mode Image Formation, Image Reconstruction and Post Image Processing. The last of these is the subject of this report, and its position at the end of the chain is shown in Figure 1.1.

2.2 Interface Specification

A precise interface contract was agreed with the upstream team so the two modules could be developed independently. Table 2.1 states it.

Table 2.1: Data interface between the B-Mode Image Formation module and the Post Image Processing module.

Property	Input (received)	Output (returned)
Container	NumPy .npz file	NumPy .npz file
Data type	uint8, 2-D array	uint8, 2-D array
Array shape	$H \times W$ (nominally 560×880)	identical to input
Value range	0–255, log-compressed	0–255, display-ready
Geometry	Cartesian, sector (fan) format	identical geometry
Auxiliary data	mask.npz — boolean array marking valid fan pixels	—
Metadata	Dynamic range D in dB, sector angle, imaging depth	unchanged

The module exposes a single entry point:

```
1 processed = post_process(raw_image, mask=mask, method="srad")
```

Listing 2.1: Public interface of the Post Image Processing module.

The mask argument is required because the fan-shaped valid region is surrounded by zero-valued padding; neighbourhood operators must not average genuine data with this padding, which would introduce a dark halo along the sector boundary.

2.3 Proposed Processing Pipeline

The module implements a four-stage pipeline: depth gain normalisation, despeckle filtering, contrast and edge restoration, and automated quality assessment.

Figure 2.1 gives the algorithmic flow, including the quality-driven parameter-selection loop.

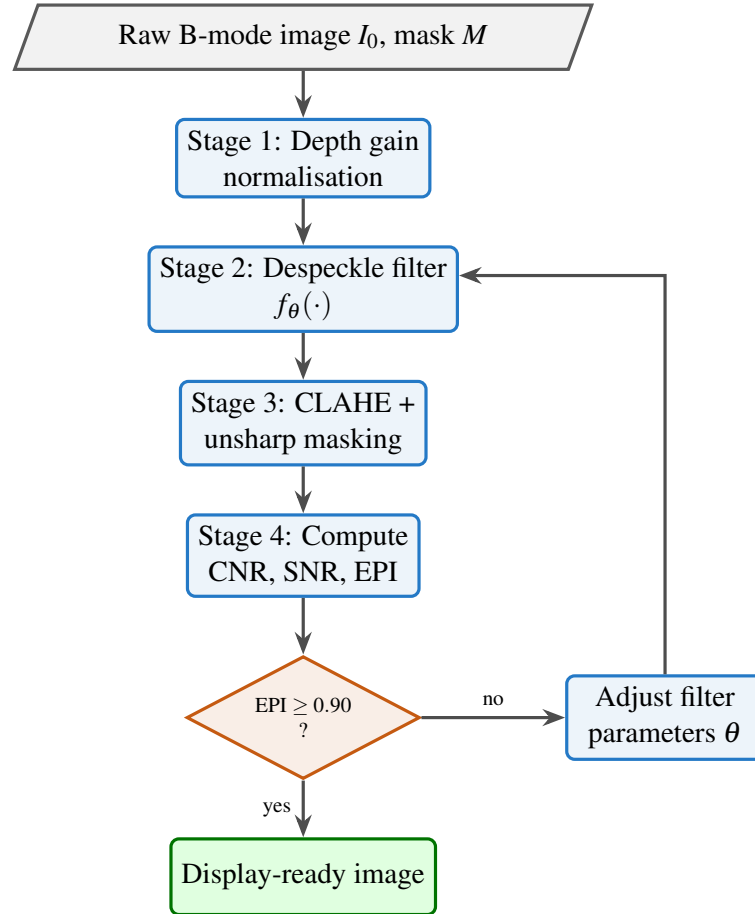


Figure 2.1: Algorithmic flow of the post-processing module, including the edge-preservation feedback loop used for parameter selection.

2.4 Stage 1: Depth Gain Normalisation

Acoustic attenuation in soft tissue is approximately exponential with depth and frequency, the received amplitude being scaled by $\exp(-\alpha f z)$ with $\alpha \approx 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$. Uncorrected, this darkens the lower image and biases both the local-statistics filters and the contrast metrics.

Stage 1 estimates the mean intensity of each depth row inside the valid mask and applies a smooth multiplicative correction so row means follow a constant target $\bar{I}_{\text{target}}$:

$$I_1(x, y) = I_0(x, y) \cdot \frac{\bar{I}_{\text{target}}}{\tilde{\mu}(y) + \varepsilon}, \quad \tilde{\mu}(y) = \mathcal{G}_\sigma * \mu(y), \quad (2.1)$$

where $\mu(y)$ is the masked row mean, $\mathcal{G}_\sigma$ a one-dimensional Gaussian kernel preventing the correction from tracking anatomy, and ε a small stabilising constant.

2.5 Stage 2: Despeckle Filtering

All five filters share a common sliding-neighbourhood principle: a local window is centred on each pixel and a new value computed from the pixels it contains. The methods differ only in how that value is derived.

2.5.1 Median Filter

The median filter replaces each pixel with the centre of its sorted neighbourhood,

$$I_2(x, y) = \text{median}\{I_1(x + i, y + j) : (i, j) \in \mathcal{N}_k\}, \quad (2.2)$$

where $\mathcal{N}_k$ is a $k \times k$ window ($k = 9$ here). Being an order statistic it resists a minority of extreme values, but is applied identically everywhere and therefore rounds corners and erodes small high-contrast targets.

2.5.2 Lee Filter

The Lee filter [5] combines the pixel value with the local mean, weighted by neighbourhood homogeneity. Let μ and σ^2 be the mean and variance in $\mathcal{N}_k$, and let

$$C_i = \frac{\sigma}{\mu} \quad (2.3)$$

be the local coefficient of variation. In a homogeneous region of fully developed speckle this takes the constant value $C_u = 1/1.91 \approx 0.523$ from Equation (1.4). The weight is

$$W = \max\left(0, 1 - \frac{C_u^2}{C_i^2}\right), \quad (2.4)$$

and the output is

$$I_2(x, y) = \mu + W \cdot (I_1(x, y) - \mu). \quad (2.5)$$

In a homogeneous region $C_i \approx C_u$, so $W \rightarrow 0$ and the output approaches the local mean; near an edge $C_i \gg C_u$, so $W \rightarrow 1$ and the pixel passes through essentially unaltered. A window of $k = 11$ was used.

2.5.3 Speckle Reducing Anisotropic Diffusion

SRAD [8] formulates despeckling as image evolution under a diffusion equation,

$$\frac{\partial I}{\partial t} = \text{div}[c(q) \nabla I], \quad (2.6)$$

where the conduction coefficient $c(q)$ decreases with a speckle-sensitive edge detector, the *instantaneous coefficient of variation*:

$$q = \sqrt{\frac{\frac{1}{2} (|\nabla I|/I)^2 - \frac{1}{16} (\nabla^2 I/I)^2}{[1 + \frac{1}{4} (\nabla^2 I/I)]^2}}, \quad (2.7)$$

and the diffusion coefficient is

$$c(q) = \frac{1}{1 + \frac{q^2 - q_0^2}{q_0^2(1 + q_0^2)}}, \quad (2.8)$$

in which $q_0(t)$ is the speckle scale function, estimated from a homogeneous region and decayed with iteration count.

Equation (2.7) is the essential difference from Perona–Malik diffusion: both terms are nor-

malised by I , so the detector responds to *relative* rather than absolute intensity change — the correct behaviour for multiplicative noise. Where $q \approx q_0$ the region is homogeneous speckle, $c \rightarrow 1$ and diffusion proceeds freely; where $q \gg q_0$ an edge is inferred, $c \rightarrow 0$ and diffusion is inhibited.

The discrete four-neighbour update is

$$I_{x,y}^{t+1} = I_{x,y}^t + \frac{\Delta t}{4} \sum_{d \in \{N,S,E,W\}} c_d \nabla_d I, \quad (2.9)$$

with $\nabla_d I$ the directional difference. A time step $\Delta t = 0.12$ and 400 iterations were used.

2.5.4 Non-Local Means

NLM [9] discards the assumption that useful information is spatially local; the restored value at pixel p is a weighted average over a search region Ω ,

$$I_2(p) = \frac{1}{Z(p)} \sum_{q \in \Omega} w(p,q) I_1(q), \quad Z(p) = \sum_{q \in \Omega} w(p,q), \quad (2.10)$$

where the weight depends on the Euclidean distance between the intensity *patches* $\mathcal{P}(p)$ and $\mathcal{P}(q)$ surrounding the two pixels:

$$w(p,q) = \exp\left(-\frac{\|\mathcal{P}(p) - \mathcal{P}(q)\|_2^2}{h^2}\right). \quad (2.11)$$

The parameter h controls weight decay: large h admits dissimilar patches and smooths more aggressively. A 5×5 patch, search radius 11 and $h = 0.16$ on a unit intensity scale were used.

2.5.5 Wavelet Soft Thresholding

The image is first mapped to the logarithmic domain via Equation (1.6), where the noise becomes approximately additive. A four-level Daubechies-4 decomposition is computed and soft thresholding [10] applied to the detail coefficients:

$$\eta_\tau(c) = \text{sign}(c) \max(|c| - \tau, 0). \quad (2.12)$$

The image is reconstructed and exponentiated back. The threshold τ derives from a robust noise estimate based on the median absolute deviation of the finest detail sub-band.

2.6 Stage 3: Contrast and Edge Restoration

Despeckling inevitably reduces local variance, and the resulting image appears flat. Two operators restore perceptual quality.

CLAHE [12] divides the image into tiles, equalises each histogram independently and interpolates between tiles to remove block artefacts. The contrast limit clips the histogram beforehand, preventing noise amplification in near-uniform tiles. Clip limit 1.3, tile grid 8×8 .

Unsharp masking recovers edge definition by subtracting a blurred copy:

$$I_3 = (1 + \lambda)I_2 - \lambda (\mathcal{G}_\sigma * I_2), \quad (2.13)$$

with $\lambda = 0.28$ and $\sigma = 2.0$ pixels.

The ordering is not arbitrary: both operators amplify high-frequency content, so applied before despeckling they would amplify the speckle itself. Despeckling always precedes enhancement.

2.7 Stage 4: Quality Assessment

Visual inspection is insufficient for parameter selection, since the same image may appear either pleasingly smooth or plastic depending on the observer. Four metrics are computed automatically.

Contrast-to-Noise Ratio. For a region of interest inside a lesion and one in adjacent normal tissue,

$$\text{CNR} = \frac{|\mu_\ell - \mu_b|}{\sqrt{\frac{1}{2}(\sigma_\ell^2 + \sigma_b^2)}}, \quad (2.14)$$

where μ_ℓ, σ_ℓ and μ_b, σ_b are the lesion and background statistics. Higher CNR indicates better lesion conspicuity.

Speckle Signal-to-Noise Ratio. Measured in a homogeneous background region,

$$\text{SNR}_{\text{speckle}} = \frac{\mu_b}{\sigma_b}. \quad (2.15)$$

By Equation (1.4) the raw value should be near 1.91 before log compression; any increase quantifies speckle suppression.

Edge Preservation Index. The correlation between the Laplacian of the processed image and that of the ground-truth image,

$$\text{EPI} = \frac{\sum (\Delta I_p - \overline{\Delta I_p})(\Delta I_g - \overline{\Delta I_g})}{\sqrt{\sum (\Delta I_p - \overline{\Delta I_p})^2 \sum (\Delta I_g - \overline{\Delta I_g})^2}}, \quad (2.16)$$

where Δ is the Laplacian, I_p the processed image and I_g the ground truth. An EPI approaching unity indicates boundary information was retained.

Structural Similarity. SSIM [13] compares luminance, contrast and structure between the processed image and the ground truth over local windows, returning unity for a perfect match.

CNR and $\text{SNR}_{\text{speckle}}$ are reference-free and measurable on clinical data; EPI and SSIM require the noise-free reference and exist only because the input comes from a simulator — exactly the advantage identified in [2]. The lesion and background ROIs are circular, of equal radius, and placed at equal depth so that residual attenuation does not bias the comparison.

2.8 Software Environment and Module Structure

The module is implemented in Python 3, chosen for the maturity of its scientific stack and consistency with the other NITK-USoundSim modules. Array arithmetic and the Lee and

SRAD implementations use NumPy [14]. SciPy provides `median_filter`, `uniform_filter` and `map_coordinates`; scikit-image [15] supplies `denoise_nl_means`, `denoise_wavelet` and the SSIM and PSNR metrics; OpenCV provides `createCLAHE` and the Gaussian blur used for unsharp masking; PyWavelets performs the Daubechies decomposition; and Matplotlib generates the result figures.

The code is organised into five source files with a single public entry point (Figure 2.2). Each stage is isolated so filters can be substituted without altering the calling code.

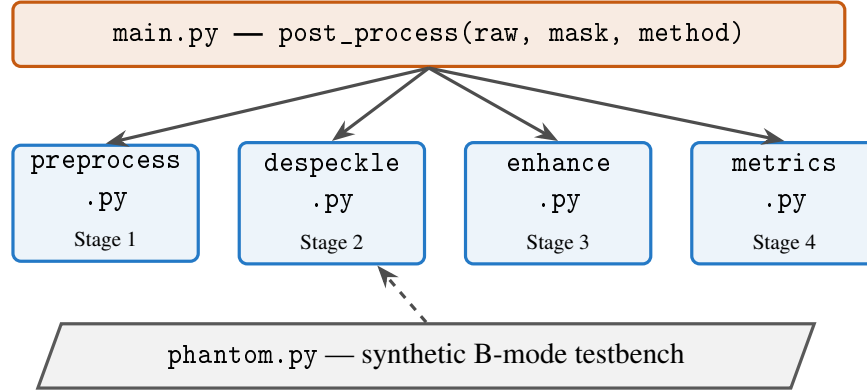


Figure 2.2: Software architecture of the Post Image Processing module. Each pipeline stage is a separate source file; the testbench supplies validation input.

- `preprocess.py` — depth gain normalisation of Equation (2.1) and dynamic-range verification.
- `despeckle.py` — the five filter functions, each with the signature `filter(image, **params)` returning an image.
- `enhance.py` — CLAHE and unsharp masking.
- `metrics.py` — CNR, $\text{SNR}_{\text{speckle}}$, EPI, SSIM and PSNR.
- `main.py` — orchestration, batch evaluation and result export.

2.8.1 Coding Pattern

Local statistics are obtained from sliding-window passes rather than explicit pixel loops, so every filter is fully vectorised. The Lee filter, for example, computes the local mean with one uniform-filter pass and the local variance from a second pass over the squared image, forms the weight of Equation (2.4) elementwise, and returns $\mu + W(x - \mu)$ — seven array operations in total, with no Python-level iteration. SRAD is the only genuinely iterative filter: its update loop implements Equations (2.7)–(2.9) using four shifted copies of the image array, so each of the 400 iterations costs a fixed number of vectorised operations. The speckle scale is decayed as $q_0(t) = 0.9/\sqrt{1 + 0.012t}$, progressively tightening the edge criterion as the image becomes smoother.

2.9 Synthetic B-Mode Testbench

Rationale

The B-Mode Image Formation module had not yet delivered production output, so a synthetic testbench was written reproducing the statistical and geometric properties of Table 2.1. Following the virtual-trial methodology of [2], the testbench also produces a noise-free reference, enabling metrics unobtainable from clinical data.

Phantom Construction

A reflectivity map $R(r, \theta)$ is defined in polar coordinates containing a bright near-field layer for skin and subcutaneous tissue, a low-amplitude sinusoidal depth variation representing tissue layering, an anechoic cyst ($R = 0.04$) as the primary low-contrast target, a hyperechoic lesion ($R = 4.5$), a hypoechoic nodule ($R = 0.12$) at greater depth, and three point targets ($R = 5.5$) for resolution assessment.

Speckle Synthesis

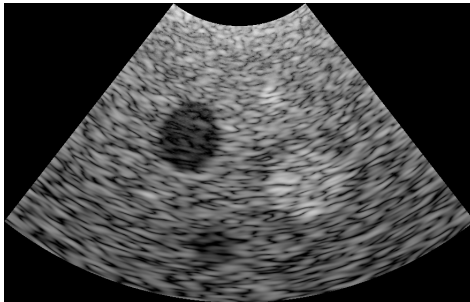
A complex circular Gaussian scattering field is modulated by the square root of the reflectivity and convolved with an anisotropic point spread function:

$$s(r, \theta) = \left[(\xi_{\text{re}} + j\xi_{\text{im}}) \sqrt{R(r, \theta)} \right] * h_{\text{PSF}}, \quad \xi \sim \mathcal{N}(0, 1), \quad (2.17)$$

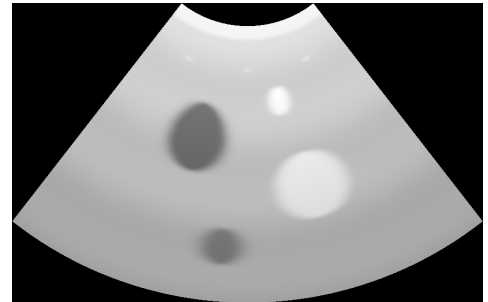
after which the envelope $E = |s|$ follows Equation (1.3). The PSF is a separable Gaussian with standard deviations 2.6 axially and 3.4 laterally, reflecting the poorer lateral resolution of ultrasound. Attenuation $\exp(-1.15r)$, log compression with $D = 40$ dB (Equation (1.2)) and scan conversion follow.

Testbench Output

Figure 2.3 shows the resulting pair. The simulated image exhibits the expected fan geometry, granular texture and depth-dependent brightness of a real sector scan. Only the anechoic cyst is clearly visible; the hypoechoic nodule and one point target are substantially masked — precisely the problem the module addresses.



(a) Simulated B-mode image with speckle



(b) Noise-free ground-truth reference

Figure 2.3: Output of the synthetic testbench. The ground-truth image (b) is available only because the input originates from a simulator, and it enables the reference-based metrics EPI and SSIM.

The testbench operates on a 520×220 polar grid rendered to a 560×880 display image over a 76° sector, with a separable Gaussian PSF ($\sigma_{\text{ax}} = 2.6$, $\sigma_{\text{lat}} = 3.4$ samples), attenuation $\exp(-1.15r)$, a 40 dB displayed dynamic range and bilinear scan conversion.

2.9.1 Experimental Protocol

A fixed protocol ensured fair comparison. A single testbench image is generated with a fixed random seed so every filter sees identical input; Stage 1 normalisation is applied once and its output passed to all filters; each filter is tuned by sweeping its principal control variable and selecting the setting with highest CNR subject to $\text{EPI} \geq 0.90$ where achievable; the mask is reapplied after filtering so the fan boundary is not contaminated by padding; and all metrics are computed against identical ROIs and the same reference.

The final parameter values are recorded in Table 2.2.

Table 2.2: Final filter parameters used for the reported results.

Filter	Parameters
Median	Window size $k = 9$
Lee	Window size $k = 11$, $C_u = 0.523$
SRAD	400 iterations, $\Delta t = 0.12$, $q_0(t) = 0.9/\sqrt{1+0.012t}$
NLM	Patch 5×5 , search radius 11, $h = 0.16$
Wavelet	Daubechies-4, 4 levels, soft threshold, $\sigma = 0.10$ in log domain
CLAHE	Clip limit 1.3, tile grid 8×8
Unsharp	$\lambda = 0.28$, $\sigma = 2.0$ pixels

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Qualitative Results

Figure 3.2 shows all five filter outputs alongside the unprocessed input, displayed with identical grey-level mapping.

The median filter reduces grain but rounds lesion boundaries and attenuates the point targets, while the Lee filter preserves boundary definition noticeably better at comparable suppression. NLM gives the smoothest tissue of any method, though fine texture is lost. Wavelet thresholding retains texture but suppresses the least speckle. SRAD, enlarged in Figure 3.1, strongly smooths homogeneous tissue while keeping cyst and lesion boundaries sharp; the hypoechoic nodule, barely visible in the input, becomes clearly discernible.

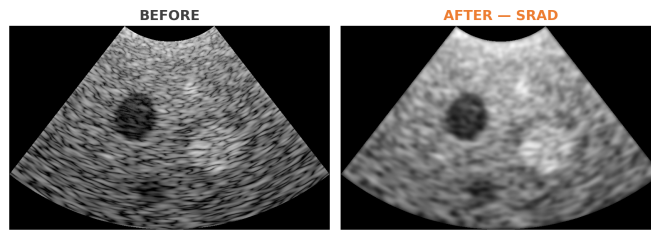


Figure 3.1: SRAD, 400 iterations: input (left) and output (right). Homogeneous tissue is strongly smoothed while the cyst and lesion boundaries remain sharp.

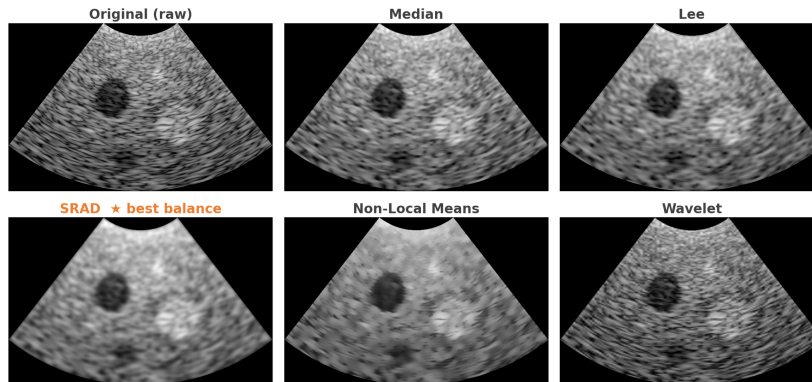


Figure 3.2: Side-by-side comparison of all five despeckling filters applied to the same input image under identical display settings.

3.2 Quantitative Results

Table 3.1 reports the measured metrics, obtained on the testbench image of Figure 2.3a against the reference of Figure 2.3b.

Figure 3.3 presents the same data graphically.

widthwidth...

Figure 3.3: Measured contrast, speckle suppression and edge preservation for each filter. The grey bar in each panel is the unprocessed reference.

Table 3.1: Measured image-quality metrics for all evaluated methods. Higher is better in every column. The best value in each column is shown in bold.

Method	CNR	SNR _{spk}	EPI	SSIM	Speed
Original (raw)	1.78	4.28	1.000*	0.484	—
Median	2.45	7.07	0.759	0.625	Very fast
Lee	2.58	7.55	0.943	0.667	Fast
Wavelet	2.00	5.08	0.570	0.509	Fast
NLM	3.03	9.77	0.806	0.728	Slow
SRAD	2.98	9.29	0.955	0.744	Moderate
SRAD + CLAHE	2.27	5.02	0.827	0.640	Moderate

*Reference value by definition; the raw image defines the unfiltered edge content.

3.3 Discussion

3.3.1 Overall Performance

Every filter improved both CNR and SNR_{speckle}. The spread is substantial: CNR gains range from 12% (wavelet) to 71% (NLM), and speckle SNR gains from 19% to 128%. The raw speckle SNR of 4.28 exceeds the theoretical 1.91 of Equation (1.4) because the metric is computed after log compression, which reduces the relative standard deviation; it remains valid as a *relative* indicator of smoothing.

3.3.2 The Contrast–Edge Trade-Off

The central finding is visible in the last two rows of the main block of Table 3.1. NLM achieves the highest CNR (3.03) and speckle suppression (9.77), but its EPI is only 0.806. SRAD attains essentially the same noise performance — CNR 2.98, a difference under 2% — while retaining an EPI of 0.955, some 18 percentage points higher, and the highest SSIM (0.744), indicating that its output is structurally closest to the true anatomy. This is the quantitative expression of the trade-off identified in Section 1.3.

Figure 3.4 demonstrates the failure mode. When smoothing strength is increased well beyond the selected operating point, the hyperechoic lesion — a genuine finding — is almost entirely erased. This is why the EPI criterion was built into the parameter-selection loop of Figure 2.1.

3.3.3 Behaviour of Individual Filters

The **median** filter shows the lowest EPI (0.759), consistent with theory: as a fixed order statistic it applies the same operation regardless of local content. The **Lee** filter performs markedly better on edges (EPI = 0.943) at similar noise suppression, validating the adaptive weighting of Equation (2.4); its weakness is that the constant- C_u assumption holds only for fully developed speckle. **SRAD** performs best overall, as expected of the only method explicitly derived for multiplicative speckle, its cost being the 400 iterations that make it roughly an order of magnitude slower than Lee. **NLM** gives the smoothest tissue, but its patch-similarity weighting is content-agnostic near boundaries where few similar patches exist, causing the observed edge softening; it was also the slowest method evaluated. **Wavelet** thresholding pre-

served texture but removed the least speckle: a more aggressive threshold raised the speckle SNR yet introduced ringing near high-contrast boundaries, so the conservative setting was retained.



Figure 3.4: Consequence of over-smoothing. At excessive filter strength the hyperechoic lesion (circled) is almost completely removed, despite the image appearing subjectively cleaner.

3.3.4 Effect of Contrast Enhancement

The last row of Table 3.1 records an important negative result. Applying CLAHE and unsharp masking after SRAD *reduced* CNR from 2.98 to 2.27 and speckle SNR from 9.29 to 5.02, although the resulting image appears subjectively crisper.

The explanation follows from Equations (2.14) and (2.15): both operators amplify local variance, and since σ_ℓ and σ_b appear in the denominators, any such increase depresses the scores even when the mean separation $|\mu_\ell - \mu_b|$ is unchanged. Stage 3 must therefore be treated as a *display-only* operation, with Stage 4 metrics computed on the Stage 2 output and enhancement applied afterwards purely for presentation.

3.3.5 Summary of Findings

1. All five classical filters suppress speckle measurably; no machine learning is required for substantial improvement.
2. SRAD offers the best balance, nearly doubling CNR ($1.78 \rightarrow 2.98$) and more than doubling speckle SNR ($4.28 \rightarrow 9.29$) while retaining 95.5% of edge information and achieving the highest SSIM.
3. NLM marginally exceeds SRAD in noise suppression, but at significant cost in edge fidelity and computation time.
4. Contrast enhancement improves subjective appearance yet degrades objective metrics, so it is excluded from the measurement path.
5. **SRAD with the parameters of Table 2.2 is the recommended default filter.**

CHAPTER 4

CONCLUSION AND FUTURE WORK

4.1 Conclusion

This report has presented the design, implementation and evaluation of the Post Image Processing module of NITK-USoundSim, which converts the raw log-compressed, scan-converted B-mode image into a display-ready image with suppressed speckle.

Speckle was shown to be a deterministic consequence of coherent imaging rather than additive electronic noise, and therefore not removable by improved hardware. The Rayleigh envelope statistics of fully developed speckle were used both to derive the Lee constant $C_u = 0.523$ and to define the speckle SNR metric.

A four-stage pipeline was designed — depth gain normalisation, despeckle filtering, contrast restoration and quality assessment — with a documented interface allowing independent development. Five classical filters were implemented in Python and compared under an identical protocol on a synthetic testbench supplying a ground-truth reference.

The measured results identify SRAD as the preferred operator. It raised the contrast-to-noise ratio from 1.78 to 2.98 and the speckle signal-to-noise ratio from 4.28 to 9.29, while retaining an edge-preservation index of 0.955 and achieving the highest structural similarity of any method tested (0.744). Non-Local Means achieved marginally better noise suppression but lost substantially more edge information and was considerably slower.

A secondary finding is that contrast enhancement, while improving subjective appearance, systematically depresses variance-normalised metrics, and must therefore sit outside the measurement path.

4.2 Limitations

Three limitations qualify these conclusions.

First, all results were obtained on a synthetic testbench rather than on production output. The testbench reproduces the properties agreed in the interface contract, but the parameters of Table 2.2 will require re-tuning against production data.

Second, the parameters were tuned on a single phantom. Generalisation across imaging depths, sector angles and tissue types has not been established.

Third, computational cost was assessed only qualitatively. SRAD at 400 iterations and NLM are both too slow for real-time operation, restricting the present implementation to offline processing.

4.3 Future Work

Validation against production data. Repeat the evaluation on images from the upstream module and re-tune the parameters accordingly.

Adaptive parameter selection. Select filter strength automatically from an estimate of the local speckle scale, allowing the module to adapt to varying imaging conditions rather than using one fixed configuration.

Computational acceleration. The SRAD update of Equation (2.9) is a local stencil operation and is well suited to GPU execution; a parallel implementation would make real-time operation feasible.

Extension of the metric set. The generalised contrast-to-noise ratio [16] is insensitive to monotonic intensity transformations and would not exhibit the enhancement artefact of Section 5.3.5. Resolution metrics based on the full width at half maximum of the point targets would further strengthen the evaluation.

Learning-based despeckling. Convolutional and adversarial networks report performance beyond classical filters, but supervised training requires paired noisy and clean images that clinical ultrasound cannot provide. As Abadi *et al.* [2] observe, a validated simulator supplies exactly such paired data at scale; the testbench developed here already generates the required references.

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